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SOC Alert Monitoring & Log Correlation

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Agenda

- Executive Abstract & Goals
- Lab Architecture & Investigation Flow
- SIEM Platform & Data Collection
- Phase 1: SMB Brute Force Detection
- Phase 2: Privilege Escalation
- Phase 3: Defense Evasion via PowerShell
- Phase 4: File Integrity Monitoring (FIM)
- Impact Analysis & Tuning Recommendations
- Conclusion & References

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Abstract & Project Goal

•**Objective:** To design and implement a step-by-step investigation workflow demonstrating how SOC analysts detect and triage suspicious behavior using log correlation.

•**Methodology:** Simulated a targeted attack lifecycle—including SMB brute-forcing, unauthorized user creation, privilege escalation, and defense evasion—and monitored the resulting telemetry.

•**Final Output:** A comprehensive Proof of Concept (PoC) mapping local Windows Security Event logs directly to Wazuh SIEM alerts, concluding with actionable recommendations for rule tuning and reducing SOC alert fatigue.

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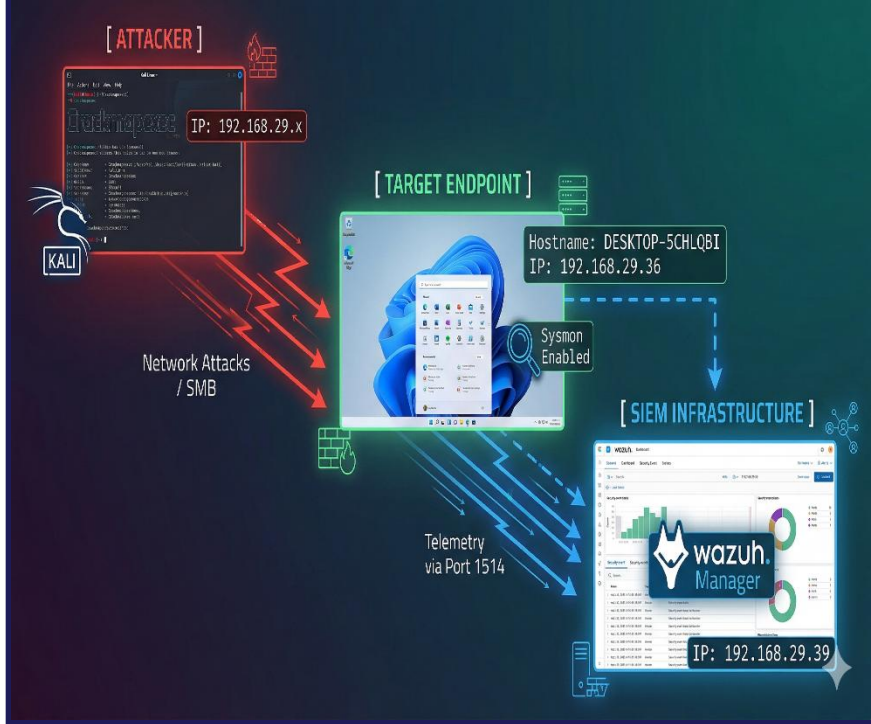
Attack Simulation Environment

To demonstrate the SOC investigation workflow, an isolated lab environment was established:

- **Attacker Machine:** Kali Linux (IP: 192.168.29.x) utilizing tools like crackmapexec to simulate external threat actor behaviors.
- **Target Endpoint:** Windows 11 Desktop (Hostname: DESKTOP-5CHLQBI) configured with advanced security auditing and Sysmon.
- **SIEM Infrastructure:** Wazuh Manager (IP: 192.168.29.39) receiving real-time telemetry from the Wazuh Agent installed on the Windows target.



Lab Environment & Architecture

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The Kill Chain & Investigation Timeline

Investigation Flow

- **Phase 1: Initial Access** → Attacker executes SMB Brute Force. *SOC Action: Investigate Event 4625 anomalous clusters.*

- **Phase 2: Persistence** → Attacker creates rogue hacker account & elevates to Admin. *SOC Action: Correlate Event 4720 & 4732.*

- **Phase 3: Evasion** → Attacker bypasses PowerShell execution policies. *SOC Action: Analyze Event 4688 with bypass flags.*

- **Phase 4: Impact** → Attacker manipulates files in C:\Users\admin\Hackers. *SOC Action: Review FIM dashboard for integrity changes.*

- **Conclusion** → Isolate host, suspend compromised accounts, and tune alerting thresholds.

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SIEM Selection: Wazuh Deep Dive

- **Name & Category:** Wazuh is an open-source, enterprise-ready Security Information and Event Management (SIEM) and Extended Detection and Response (XDR) platform.
- **Tool Justification:** Wazuh was chosen over alternatives like ELK Stack or Splunk Free due to its open-source flexibility, lightweight agent architecture, and robust out-of-the-box correlation rules.
- **Popularity:** Highly popular in enterprise and academic environments due to its comprehensive MITRE ATT&CK mapping and seamless endpoint telemetry integration.

SIEM Platform Research

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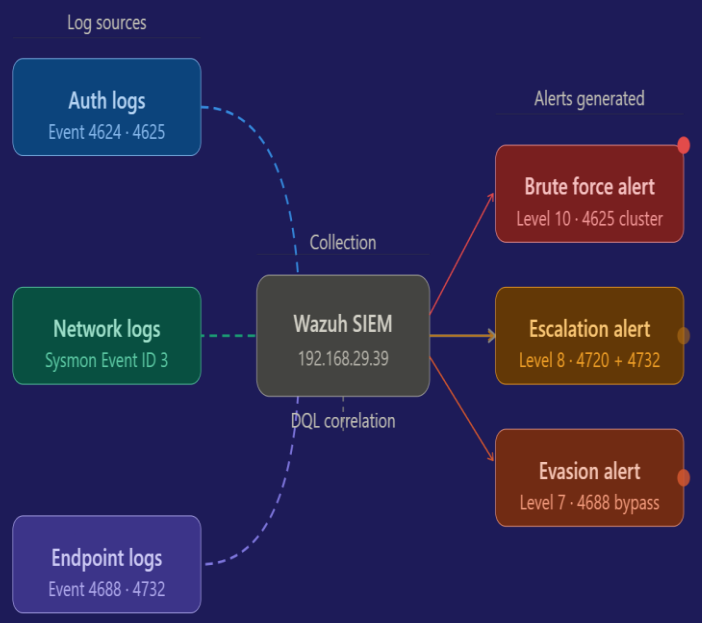
Telemetry, Data Collection & Filtering



Log Data Collection Strategy

- To ensure total visibility, the Wazuh agent collected critical endpoint telemetry. Analysts utilized Dashboard Query Language (DQL) filters (e.g., data.win.system.eventID) to isolate the attacker's footprint amidst normal background noise:
- **Authentication Logs:** Auditing Logon events (Success/Failure) to detect brute-force activity.
- **Network Logs:** Monitoring remote SMB connections originating from the attacker subnet utilizing Sysmon Event ID 3 (Network Connections) and native Windows Security logs.
- **Endpoint/System Logs:** Utilizing Windows Advanced Audit Policies to capture Process Creation (Event 4688) and Security Group Management (Event 4732).

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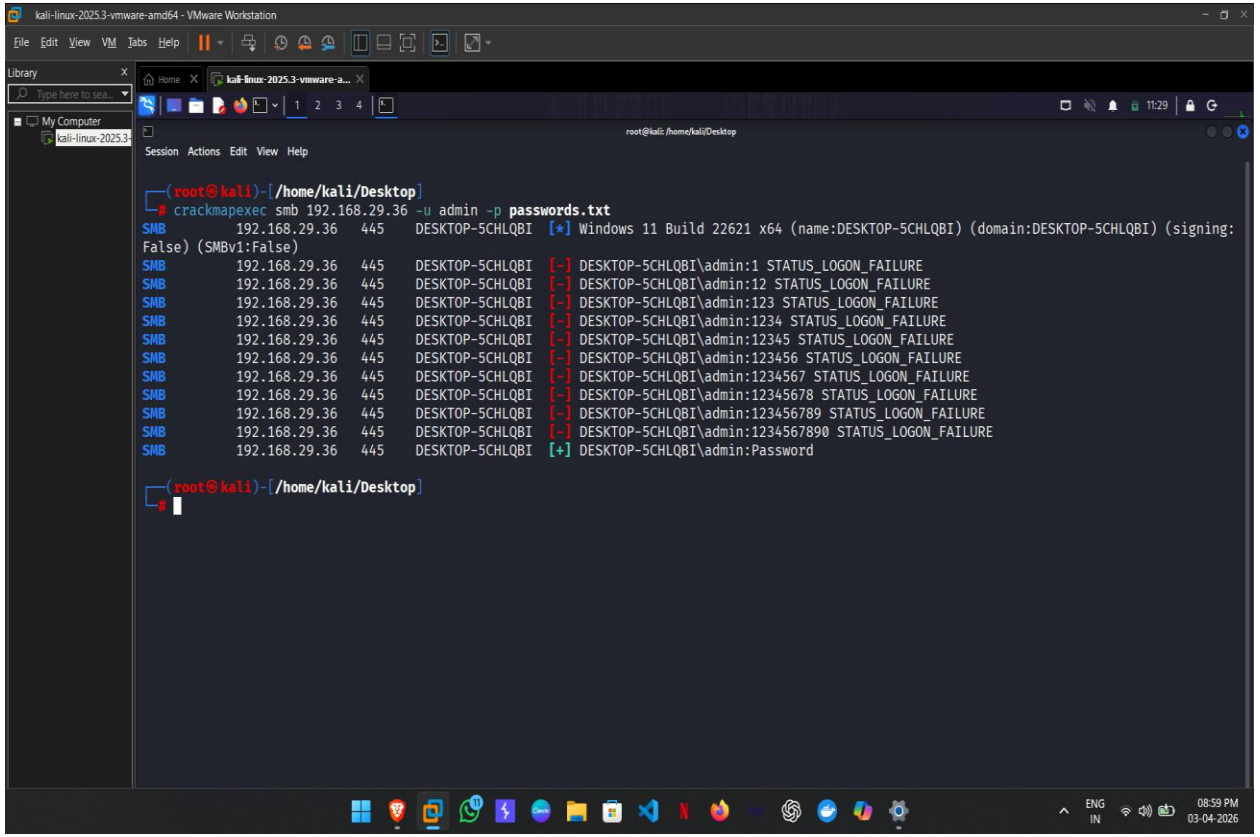
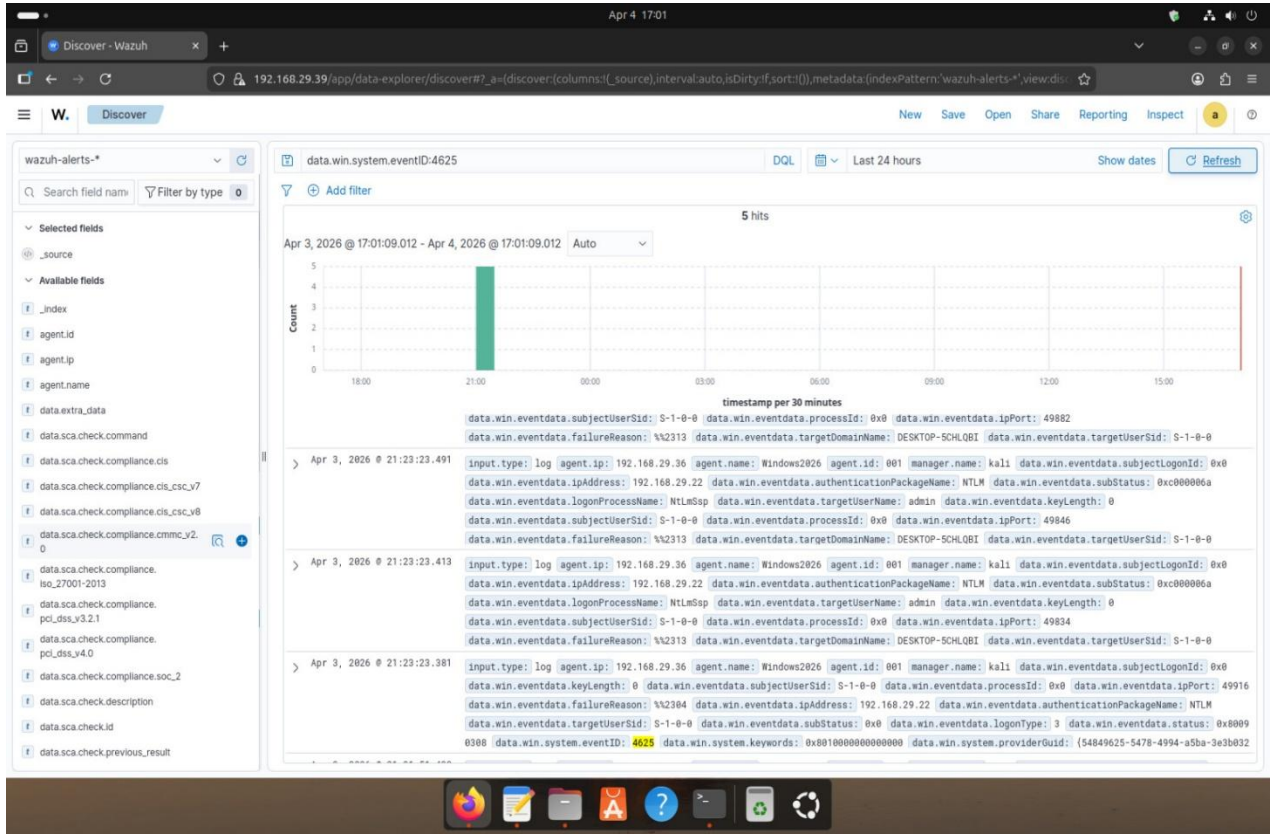
Attack Phase 1: SMB Brute Force Detection

The attack simulation commenced with a dictionary brute-force attack against the endpoint's SMB service using crackmapexec. The SIEM successfully correlated a rapid succession of **Event ID 4625** (STATUS_LOGON_FAILURE), immediately followed by a successful authentication (**Event ID 4624**).





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Phase 1 Forensic Evidence: Event ID 4625 Detail

This expanded log identifies the specific targeted account (`admin`) and the attacker's source IP address (`192.168.29.22`), providing the forensic link between the brute-force attempt and the Kali machine."

Apr 3 21:33

Discover - Wazuh | Wazuh

192.168.29.39/app/discover#/doc/wazuh-alerts-*/wazuh-alerts-4.x-2026.04.03?id=WfcMVJ0BD13OEdUTsOE5 90%

Discover | wazuh-alerts-4.x-2026.04.03#WfcMVJ0BD13OEdUTsOE5

data.win.system.channel	Security
data.win.system.computer	DESKTOP-5CHLQBI
data.win.system.eventID	4625
data.win.system.eventRecordID	657781
data.win.system.keywords	0x8010000000000000
data.win.system.level	0
data.win.system.message	

```
"An account failed to log on.

Subject:
        Security ID:            S-1-0-0
        Account Name:           -
        Account Domain:         -
        Logon ID:               0x0

Logon Type:                     3

Account For Which Logon Failed:
        Security ID:            S-1-0-0
        Account Name:           admin
        Account Domain:         DESKTOP-5CHLQBI

Failure Information:
        Failure Reason:         Unknown user name or bad password.
        Status:                 0xC000006D
        Sub Status:             0xC000006A

Process Information:
        Caller Process ID:      0x0
        Caller Process Name:    -

Network Information:
        Workstation Name:       -
        Source Network Address: 192.168.29.22
        Source Port:            49882

Detailed Authentication Information:
        Logon Process:          NtLmSsp
        Authentication Package: NTLM
        Transited Services:     -
        Package Name (NTLM only):       -
        Key Length:             0

This event is generated when a logon request fails. It is generated on the computer where access was attempted.
```

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Attack Phase 2: Privilege Escalation & Persistence

After gaining a foothold, the attacker successfully escalated privileges to create a rogue user (hacker) (**Event ID 4720**) and added it to the local Administrators group. The SIEM captured **Event ID 4732**, triggering a critical severity alert for unauthorized privilege escalation.

Administrator: Command Prompt

```
C:\Users\admin>net localgroup administrators hacker /add
The command completed successfully.


C:\Users\admin>net localgroup administrators
Alias name     administrators
Comment        Administrators have complete and unrestricted access to the computer/domain

Members

-------------------------------------------------------------------------------
admin
Administrator
hacker
The command completed successfully.


C:\Users\admin>_
```

Event Viewer

File Action View Help

Event Viewer (Local)
- Custom Views
- Windows Logs
  - Application
  - Security
  - Setup
  - System
  - Forwarded Events
- Applications and Services Lo
- Subscriptions

Security Number of events: 30,325

Filtered: Log: Security; Source: ; Event ID: 4732. Number of events: 2

Keywor...	Date and Time	Source	Event ID	Task Ca...
Audi...	04/04/2026 12:24:06	Micros...	4732	Securit...
Audi...	04/04/2026 12:18:58	Micros...	4732	Securit...

Event 4732, Microsoft Windows security auditing.

General Details

A member was added to a security-enabled local group.

Subject:
- Security ID: DESKTOP-5CHLQBI\admin
- Account Name: admin
- Account Domain: DESKTOP-5CHLQBI
- Logon ID: 0x3E2ED

Member:
- Security ID: DESKTOP-5CHLQBI\hacker
- Account Name: -

Group:
- Security ID: BUILTIN\Administrators
- Group Name: Administrators
- Group Domain: Builtin

Additional Information:
- Privileges: -

Log Name:	Security		
Source:	Microsoft Windows security	Logged:	04/04/2026 12:24:06
Event ID:	4732	Task Category:	Security Group Management
Level:	Information	Keywords:	Audit Success
User:	N/A	Computer:	DESKTOP-5CHLQBI
OpCode:	Info		
More Information:	Event Log Online Help		

Actions

Security
- Open Saved Log...
- Create Custom View...
- Import Custom View...
- Clear Log...
- Filter Current Log...
- Clear Filter
- Properties
- Find...
- Save Filtered Log File As...
- Attach a Task To this Log...
- Save Filter to Custom View...
- View
- Refresh
- Help

Event 4732, Microsoft Windows security auditing.
- Event Properties
- Attach Task To This Event...
- Copy
- Save Selected Events...
- Refresh
- Help

Creates a filter.

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SIEM Correlation: Privilege Escalation

The local endpoint events were immediately ingested by Wazuh. The SIEM successfully correlated the creation of the rogue account (Event 4720) and the critical modification of the built-in Administrators group (Event 4732), triggering high-severity alerts.

Apr 4 15:49

Discover - Wazuh | Wazuh

192.168.29.39/app/discover#/doc/wazuh-alerts-*/wazuh-alerts-4.x-2026.04.04?id=6H0AWJ0BqRFg2tQko-Hs

Discover | wazuh-alerts-4.x-2026.04.04#6H0AWJ0BqRFg2tQko-Hs

data.win.system.eventID	4720
data.win.system.eventRecordID	698877
data.win.system.keywords	0x8020000000000000
data.win.system.level	0
data.win.system.message	"A user account was created.

```
Subject:
        Security ID:            S-1-5-21-3298235045-1542125364-3046189717-1001
        Account Name:           admin
        Account Domain:         DESKTOP-5CHLQBI
        Logon ID:               0x3E2ED

New Account:
        Security ID:            S-1-5-21-3298235045-1542125364-3046189717-1020
        Account Name:           hacker
        Account Domain:         DESKTOP-5CHLQBI

Attributes:
        SAM Account Name:       hacker
        Display Name:           <value not set>
        User Principal Name:    -
        Home Directory:         <value not set>
        Home Drive:             <value not set>
        Script Path:            <value not set>
        Profile Path:           <value not set>
        User Workstations:      <value not set>
        Password Last Set:      <never>
        Account Expires:                <never>
        Primary Group ID:       513
        Allowed To Delegate To: -
        Old UAC Value:          0x0
        New UAC Value:          0x15
        User Account Control:
                Account Disabled
                'Password Not Required' - Enabled
                'Normal Account' - Enabled
        User Parameters:        <value not set>
        SID History:            -
        Logon Hours:            All

Additional Information:
```

Apr 4 16:12

Discover - Wazuh | Wazuh | (9) WhatsApp

192.168.29.39/app/discover#/doc/wazuh-alerts-*/wazuh-alerts-4.x-2026.04.04?id=UX0VWJ0BqRFg2tQkHuOL

Discover | wazuh-alerts-4.x-2026.04.04#UX0VWJ0BqRFg2tQkHuOL

data.win.eventdata.subjectLogonId	0x3e2ed
data.win.eventdata.subjectUserName	admin
data.win.eventdata.subjectUserSid	S-1-5-21-3298235045-1542125364-3046189717-1001
data.win.eventdata.targetDomainName	Builtin
data.win.eventdata.targetSid	S-1-5-32-544
data.win.eventdata.targetUserName	Administrators
data.win.system.channel	Security
data.win.system.computer	DESKTOP-5CHLQBI
data.win.system.eventID	4732
data.win.system.eventRecordID	716051
data.win.system.keywords	0x8020000000000000
data.win.system.level	0
data.win.system.message	"A member was added to a security-enabled local group.

```
Subject:
        Security ID:            S-1-5-21-3298235045-1542125364-3046189717-1001
        Account Name:           admin
        Account Domain:         DESKTOP-5CHLQBI
        Logon ID:               0x3E2ED

Member:
        Security ID:            S-1-5-21-3298235045-1542125364-3046189717-1020
        Account Name:           -

Group:
        Security ID:            S-1-5-32-544
        Group Name:             Administrators
        Group Domain:           Builtin

Additional Information:
        Privileges:             -"
```

data.win.system.opcode	0

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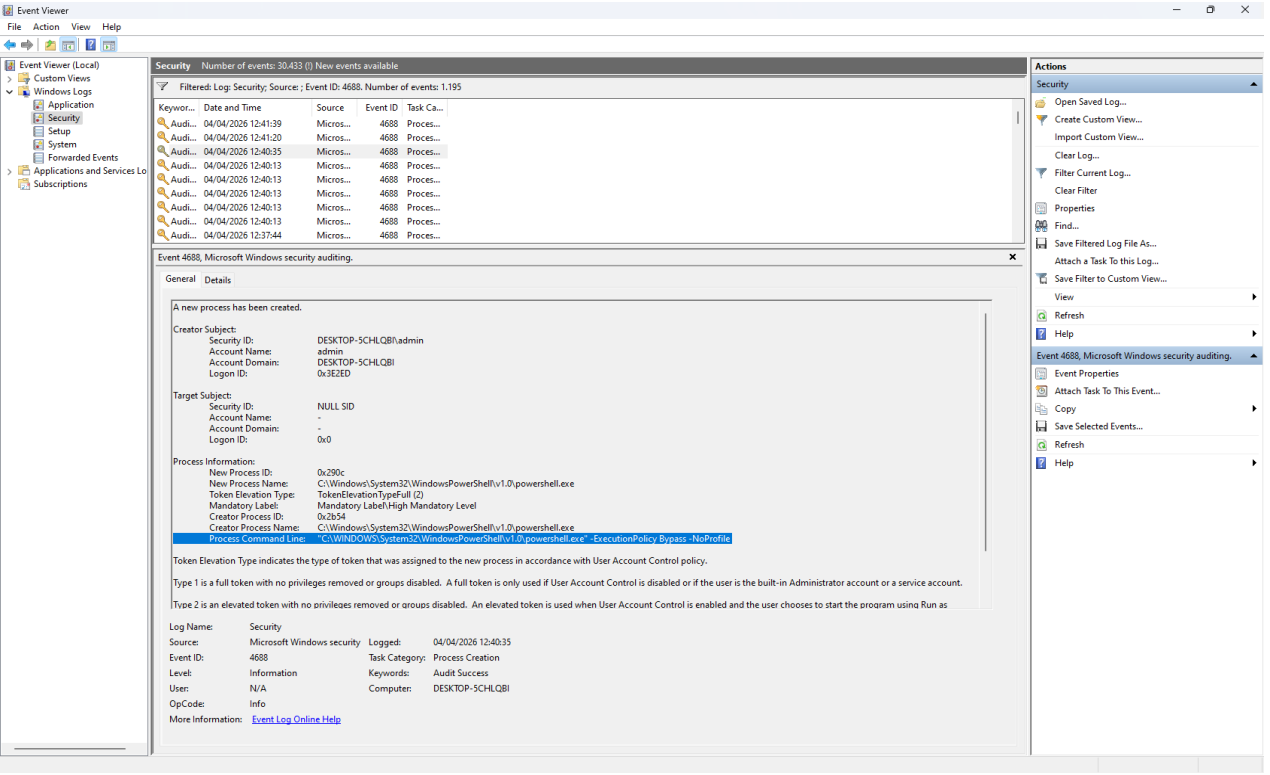
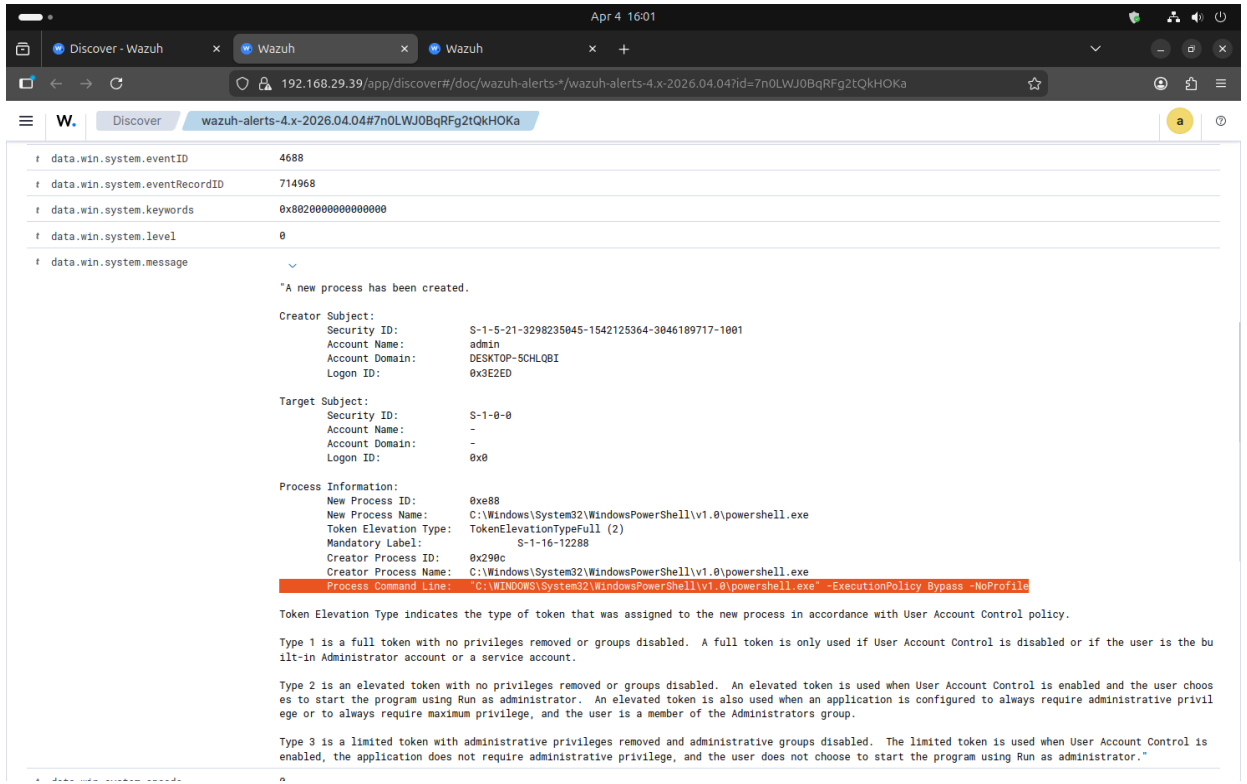
Attack Phase 3: Defense Evasion via PowerShell

To execute payloads without interference from local execution policies, the attacker utilized PowerShell bypass techniques. Command Line Auditing captured **Event ID 4688**. The SIEM successfully correlated the `powershell.exe` process with the explicit flags `-ExecutionPolicy Bypass -NoProfile`.





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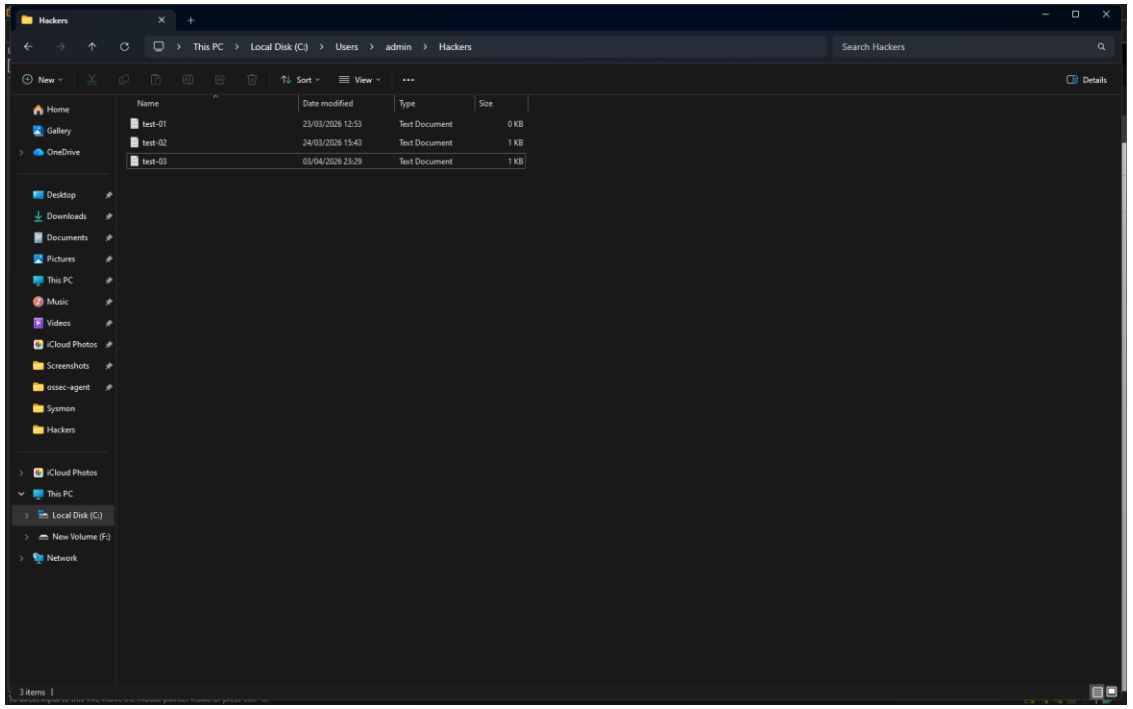
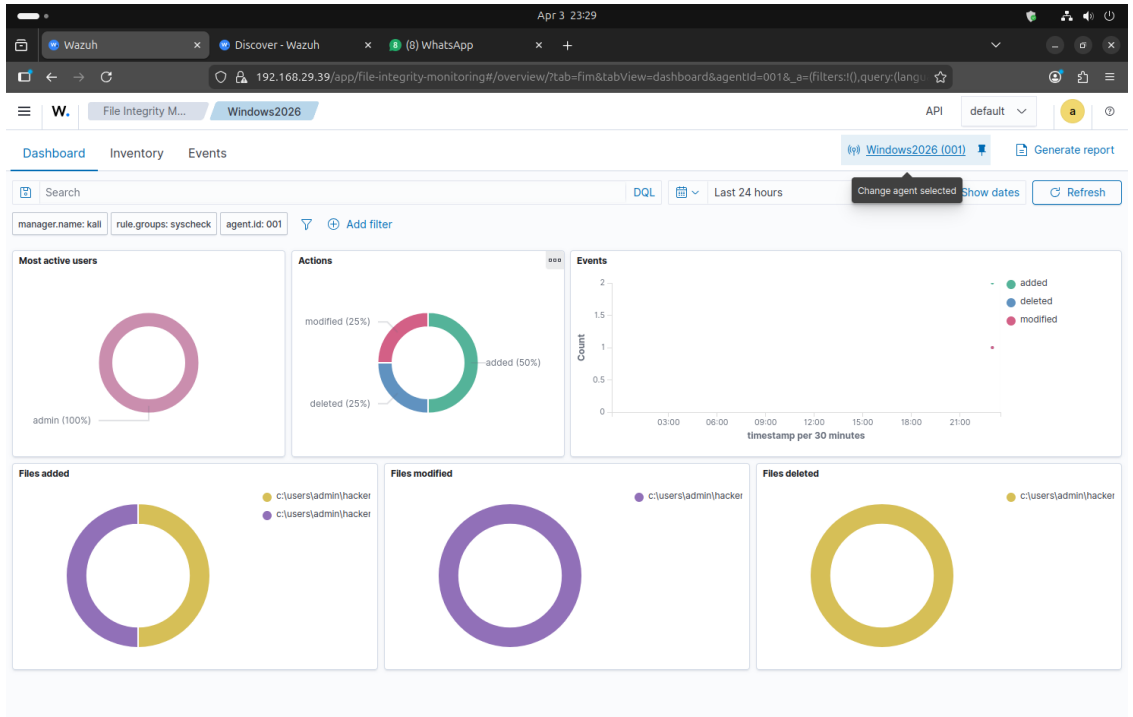
Attack Phase 4: File Integrity Monitoring (FIM)

Wazuh's File Integrity Monitoring (Syscheck) actively monitored the `C:\Users\admin\Hackers` directory. The SIEM dashboard provided a correlated timeline of the attacker adding, modifying, and deleting text files, demonstrating the immediate impact on endpoint integrity.





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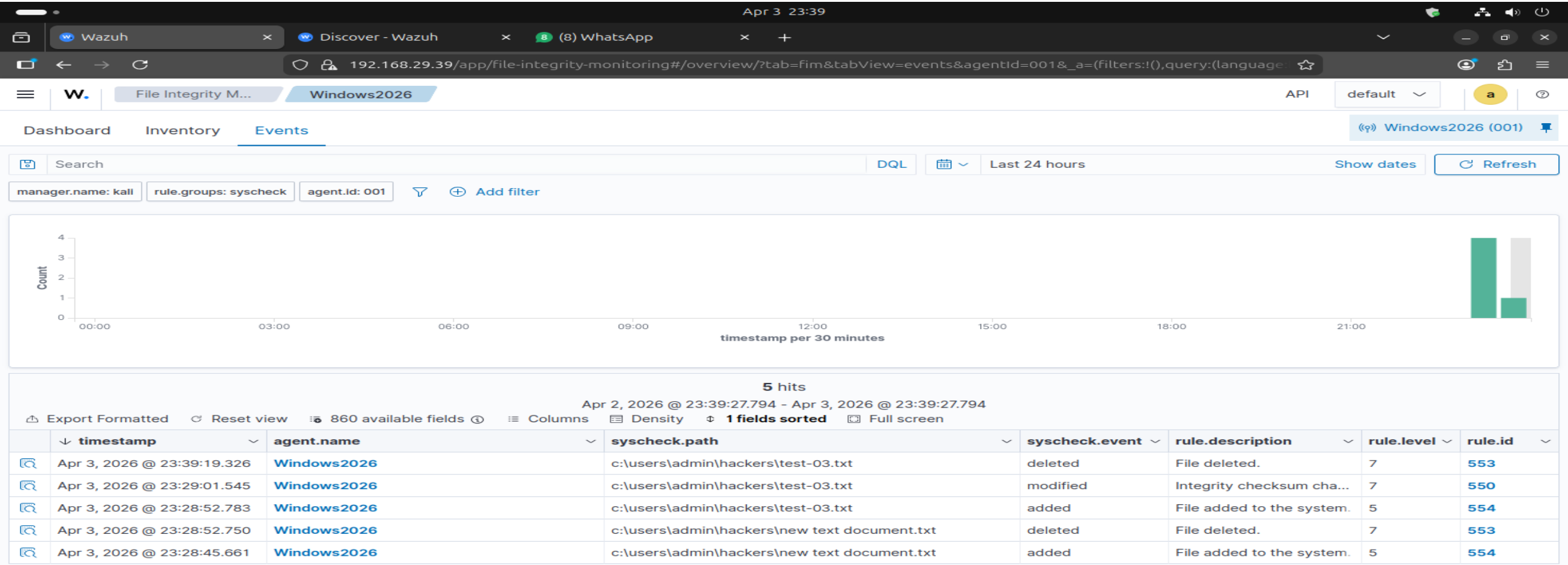
Wazuh successfully captured a detailed timeline of the file manipulations. The SIEM logged the exact timestamps and file paths (e.g., `test-03.txt`, `new text document.txt`) for every "added", "modified", and "deleted" action, providing clear forensic evidence of the attacker's impact on the system.



timestamp	agent.name	syscheck.path	syscheck.event	rule.description	rule.level	rule.id
Apr 3, 2026 @ 23:39:19.326	Windows2026	c:\users\admin\hackers\test-03.txt	deleted	File deleted.	7	553
Apr 3, 2026 @ 23:29:01.545	Windows2026	c:\users\admin\hackers\test-03.txt	modified	Integrity checksum cha...	7	550
Apr 3, 2026 @ 23:28:52.783	Windows2026	c:\users\admin\hackers\test-03.txt	added	File added to the system.	5	554
Apr 3, 2026 @ 23:28:52.750	Windows2026	c:\users\admin\hackers\new text document.txt	deleted	File deleted.	7	553
Apr 3, 2026 @ 23:28:45.661	Windows2026	c:\users\admin\hackers\new text document.txt	added	File added to the system.	5	554

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Impact Analysis of Alert Correlation

Impact Analysis

- **Contextualizing Noise:** Raw logs are noisy. An isolated failed logon (4625) has low impact. Correlating multiple failed logons to a subsequent User Creation event (4720) flags a high-impact compromise, prioritizing triage.
- **Tracing the Kill Chain:** By linking network authentication logs to local process creation logs, the SOC can determine the exact scope of the breach and prevent lateral movement.
- **Reducing Dwell Time:** Automated correlation rules reduce the time an attacker resides undetected. Rapid correlation of the Administrator group modification (4732) allows for immediate account suspension before data exfiltration.

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Tuning & False Positives

- To optimize the SIEM and reduce alert fatigue, the following measures are recommended:

- **Tuning Brute Force Alerts (4625):** Adjust the correlation threshold to trigger only if >15 failures occur within a 3-minute window from the same source IP, filtering out standard typing errors.

- **Filtering Process Creation (4688):** Create child rules that only trigger high-severity alerts when 4688 is accompanied by suspicious arguments (e.g., ep bypass, Hidden) while whitelisting known updater services.

- **FIM Exclusions:** Tune the syscheck rule to ignore highly volatile directories like C:\\Windows\\Temp to prevent dashboard flooding, focusing strictly on critical user directories and registry keys.

Alert Tuning Recommendations

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Thoughts & Project Conclusion

- Implementing a SIEM is only the first step; the true value of a SOC lies in **logical correlation**.
- By successfully mapping raw, isolated Windows Event logs to a centralized Wazuh dashboard, this project demonstrated how analysts transform raw data into actionable threat intelligence.
- Moving forward, continuous tuning of correlation rules (like FIM exclusions and Brute Force thresholds) is critical to maintaining visibility while preventing SOC analyst burnout from false positives.

Conclusion

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References

References & Methodology

•**SIEM Documentation:** Wazuh Official Documentation (Log Data Collection, Syscheck).
• **Log Sourcing:** Microsoft Learn - Windows Security Audit Events.

•**SOC Methodology (MITRE ATT&CK):**
•**T1110:** Brute Force *(Covers Phase 1: Failed and Successful Logons)*

•**T1136.001:** Create Account: Local Account *(Covers Phase 2: Persistence)*

•**T1098:** Account Manipulation *(Covers Phase 2: Privilege Escalation to Admin)*

•**T1059.001:** Command and Scripting Interpreter: PowerShell *(Covers Phase 3: Execution)*

•**T1562.001:** Impair Defenses: Disable or Modify Tools *(Covers Phase 3: Defense Evasion)*

•**T1565.001:** Data Manipulation: Stored Data Manipulation *(Covers Phase 4: FIM / File Modifying)*

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Questions ?

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Thank You!